CSE4104-7C-T04 Backend Progress Report

Project Title

Pluto AI — AI Code Security Analyzer

Team Information

- Team: CSE4104-7C-T04
- Section: 7C
- Course: CSE4104 — Software Development III
- Repository: https://github.com/0xSaikat/pluto-ai

Backend Technology Stack

- Node.js
- Express.js
- PostgreSQL (Sequelize ORM)
- JWT Authentication
- bcrypt password hashing

Database Design Summary

Implemented PostgreSQL Tables

#	Table	Description
1	users	User accounts, hashed passwords, and role assignments
2	projects	Project groups owned by individual users
3	scans	Individual scan records with status and risk score
4	source_codes	Source code content or GitHub URL per scan
5	vulnerabilities	Detected vulnerabilities with CWE ID and severity
6	reports	Auto-generated security reports stored as JSONB
7	activity_logs	Audit trail recording all user actions with timestamps
8	feedback	User ratings and comments on generated reports

Relationships are implemented using foreign keys. A user can own multiple projects and activity logs, each project can contain multiple scans, and each scan is linked to one source code entry,

multiple vulnerabilities, and one report. Reports can have multiple feedback entries. All foreign key relationships use onDelete: CASCADE to maintain referential integrity.

Implemented APIs

Authentication APIs

- POST /api/auth/register — Register new user account
- POST /api/auth/login — Login and receive JWT token
- GET /api/auth/me — Get currently authenticated user profile

User APIs

- GET /api/users — List all users (admin only)
- GET /api/users/me — Get own profile details

Project APIs

- POST /api/projects — Create a new project
- GET/api/projects — List all projects for logged-in user
- GET /api/projects/:id — Get one project with scan history
- PUT /api/projects/:id — Update project name or description
- DELETE /api/projects/:id — Delete project and all related data

Scan APIs

- POST /api/scans — Submit source code and trigger AI analysis
- GET /api/scans — List all scans for logged-in user
- GET /api/scans/:id — Get full scan result with vulnerabilities
- POST /api/scans/:id/rescan — Re-run analysis on existing scan

Vulnerability APIs

- GET /api/vulnerabilities — List all detected vulnerabilities
- GET /api/vulnerabilities/:id — Get one vulnerability with full details
- PUT /api/vulnerabilities/:id/status — Update status to open, fixed, or ignored

Report APIs

- GET /api/reports — List all generated security reports
- GET /api/reports/:id — Get full report with vulnerability summary
- GET /api/reports/:id/download — Download report as JSON file

Activity Log and Dashboard APIs

- GET /api/activity-logs — View all system logs (admin only)
- GET /api/activity-logs/me — View own activity history
- GET /api/dashboard — Get stats and recent activity for dashboard

API Testing

Postman collection exported at:

- Location: /backend/CSE4104-7C-T04_APICollection.json

Screenshots Checklist

- Database tables/collections
- API testing success/error responses
- Authentication process
- Backend folder structure
- GitHub repository activity